

Niraj Sagar Sanas

Curriculum Vitae

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Education

Fergusson College (Autonomous), Pune, India Bachelor of Science (Physics), CGPA: 8.51/10 (Overall) 9.60/10.0 (Major - Final Year)	Aug 2022 – May 2025
Swami Vivekanand Junior College, Jalgaon, India Higher Secondary Certificate, Percentage: 74.17%	Jun 2020 – May 2022
Vivekanand Pratishthan English Medium School, Jalgaon, India Secondary School Certificate, Percentage: 93.40%	Jun 2019 – May 2020

Research Experience

Determining Radio Luminosity Around Extreme Emission Line Galaxies Curtin Institute of Radio Astronomy, Perth, Australia Bachelor Thesis, Adviser: Dr. Anshu Gupta	July 2024 – May 2025
- Investigated 76 Extreme Emission Line Galaxies by extracting JWST/NIRCam cutouts from the JADES survey and MeerKAT radio data from the MIGHTEE survey to determine radio signals from AGN contributions to the energy budget of early galaxies. - Wrote Python code (using Astropy, Photutils, and Matplotlib) for image cutout extraction, WCS header recalibration, background subtraction, and flux measurement. - Used SAO DS9 and Python scripts to align infrared and radio images, identify starburst and AGN signatures, and classify fluxes $\geq 30 \mu\text{Jy}$ as statistically significant ($\geq 3\sigma$), with the majority of sources below this threshold.	

Projects

Spectral Analysis of Galaxy M100	Apr 2025
- Isolated the Hα emission by subtracting the continuum from HST dataset for extraction of nebular features. - Fitted a gaussian profile to the Hα line and measured the central wavelength and spectral width. - Calculated redshift from the wavelength shift and distance of 13.34 Mpc.	
Exoplanets Study through Kepler and Tess Transit Data	May 2024
- Identify exoplanets transits using Python libraries (Lightcurve & AstroPy) from the Kepler and TESS datasets. - Calculated the planet-to-star radius ratio and planetary size relative to its host star using transit depth. - Used Kepler's Third Law to determine orbital semi-major axes and estimate orbital distances.	

Articles & Citizen Science

Exoplanet Detection: A Review of Detection Techniques and Discoveries	Feb 2024
- Provided a concise overview of five exoplanet detection methods: Radial Velocity, Transit Photometry, Gravitational Microlensing, Direct Imaging, and Astrometry. - Highlighted discoveries such as TRAPPIST-1 atmospheric features and WASP-96b spectral detections; alongside next-generation observatories to enhance detection sensitivity and efficiency.	
Exoplanet Watch	May 2024
Contributed to NASA citizen science project, where I locate various transits of exoplanets including CoRoT-1b and WASP-36b orbiting their respective host stars.	

Technical Skills

Languages	English, Marathi, Sanskrit, Hindi
Programming languages	Python, R programming, LaTeX, C, HTML, CSS
Softwares & Tools	SAO-DS9, Origin, Stellarium, Microsoft Office, GitHub
OS	Windows, Linux

Online Courses

- **AstroTech: Science behind Astronomical Discovery** by Coursera, University of Edinburgh [Credential](#)
- **Astro 101: Black Holes** by Coursera, University of Alberta [Credential](#)
- **100 Days of Python Bootcamp**, Udemy [Credential](#)

Workshops & Schools

- **Elementary Introduction to Astrophysical Plasma Workshop** July 2024
Speaker: Dr. Nishant Singh, Inter-University Centre for Astronomy and Astrophysics, India
- **LIGO Gravitational Wave Open Data Workshop** May 2024
Inter-University Centre for Astronomy and Astrophysics, India
- **Introduction to Computational Astrophysics** Mar 2024
Speaker: Parikshit Biswas, Nicolaus Copernicus Astronomical Center

Poster Presentations

- **Frontiers in Physics XVIII**, Fergusson College, India
• Characteristic Study of Extreme Emission Line Galaxies Feb 2025
- **Astro Club**, Fergusson College, India
• Astronomical Spectroscopy and Spirit Rover model Jan 2024
• Exoplanets: Methods of Detection and Juno Spacecraft model Jan 2023
- **National Science Day**, Inter-University Center for Astronomy & Astrophysics, India
• Presented on AstroSat exhibits to an average of 2000+ people. Feb 2024
• Explained Black Hole formation and its behavior.

Achievements

- Secured 3rd prize for presenting poster on Astronomical Spectroscopy. 2023
- Secured 2nd prize for presenting poster on Exoplanets and Methods of Detection. 2022
- Ranked in the top 1% in the Secondary Board Examination. 2020
- Secured 1st prize in a Science Exhibition competition. 2019

Co-Curricular Activities

- **Astro Club**, Fergusson College, Pune 2022 – 2025
• Organized regular group discussions and physics seminars.
• Led the club's finance team, overseeing budget planning and logistical coordination for all events.
• Mentored undergraduate students in academic related courses, designing event posters and CV drafting.
- **Social Media Management & Freelancing** 2021 – Present
• Manage the [@filmwaffles](#) Instagram channel with over 11,000 followers by creating and editing video content.